

Alkaloids Questions prepared by **Mark**.

1. Which of the following alkaloid is different from the others?

- A. Ephedrine B., Colchicine C. Caffeine D.
Mescaline

2. Which of the following is Pseudo alkaloid?

- A. Ephedrine B. Colchicine C. Caffeine D.
Mescaline

3. ----- that are not derived from amino acids but have nitrogen in a heterocyclic ring

- A. Typical alkaloids B. Protoalkaloids C.
Pseudo alkaloids D. False alkaloids

4. ----- are non-alkaloids give false positive reaction with alkaloidal reagents

- A. Typical alkaloids B. Protoalkaloids C.
Pseudo alkaloids D. False alkaloids

5. Which of the following is not distribution and occurrence of alkaloids?

- A. They may be of plant, animal, insect or
microbial origin
B. Rare in lower plants
C. Monocots are richer in alkaloids than Dicots.
D. None

6. Which of the following are/is not Families rich in Alkaloids?

A. Apocynaceae B. Rubiaceae C. Solanaceae
D. Papaveracea E. None

7 Which of the following are/is Families free from Alkaloids?

A. Rosaceae B. Labiatae C. Solanaceae **D. A and B** E. None

8. Which of the following is gluco-alkaloids?

A. Solanine in Solanum B. Meconic acid in Opium
C. Quinic acid in Cinchona D. All

9. Which of the following are function of alkaloid in plants?

A. Protective and Growth regulators
B. Products of detoxification
C. Source of nitrogen and energy
D. All

10. Which of the following drug is named by discoverer?

A. Ergotamine **B. Pelletierine** C. Emetine D. Hygrine

11. Which of the following is true about prefix and suffixes?

A. "**Nor-**" designates N-demethylation or N-demethoxylation

B. "**Apo-**" designates dehydration

C. "**Iso-, pseudo-, neo-, and epi-**" indicate types of isomers

D. "-nine" indicates lower pharmacological activity
E. All

12. Which of the following are physical properties of alkaloids?

A. Most alkaloids are crystalline solids

B. Few alkaloids are amorphous solids e.g. emetine.

C. Some are liquids that are either: Volatile or Non-volatile

D. The majority of alkaloids are colorless but some are colored

E. All

13. Which of the following is Tertiary amines?

A. Norephedrine B. Ephedrine C. Atropine D. d-Tubocurarine

14. Which of the following alkaloid is not basic?

A. Primary amines B. Secondary amines C. Tertiary amines D. Quaternary amines

15. Which of the following alkaloid is act as both acidic and basic?

A. amide alkaloid B. phenolic alkaloid C. Primary alkaloid D. Quaternary alkaloid

16. Which of the following alkaloid is act as neutral?

A. amide alkaloid B. phenolic alkaloid C. Primary alkaloid D. Quaternary alkaloid E. B and C. F. A and D

17. Which of the following is mismatched according to basicity alkaloids?

A. Weak bases e.g. Caffeine

B. Strong bases e.g. Atropine

C. Amphoteric e.g. Morphine

D. Neutral alkaloids e.g. Colchicine

E. None

18. Which of the following alkaloid is solid and contain oxygen?

A. atropine B. Nicotine C. Coniine D. All

19. Alkaloids are decomposed by heat, except

A. Strychnine B. caffeine (sublimable) C. A and B D. All

20. Which of the following is true about reaction of alkaloid with acids?

A. Dilute acids hydrolyze Ester Alkaloids e.g. Atropine

B. Conc. acids may cause dehydration: e.g. Atropine → Apotatropine

C. Conc. acids may cause demethoxylation: e.g. Codeine

D. All

21. Which of the following effects of Dilute alkalis e.g. NH_3

A. liberate most alkaloids from their salts

B. may cause racemization (hyoscyamine to atropine)

C. Can form salts with alkaloids containing a carboxylic group e.g. narceine

D. All

22. Which of the following effects of strong alkalis?

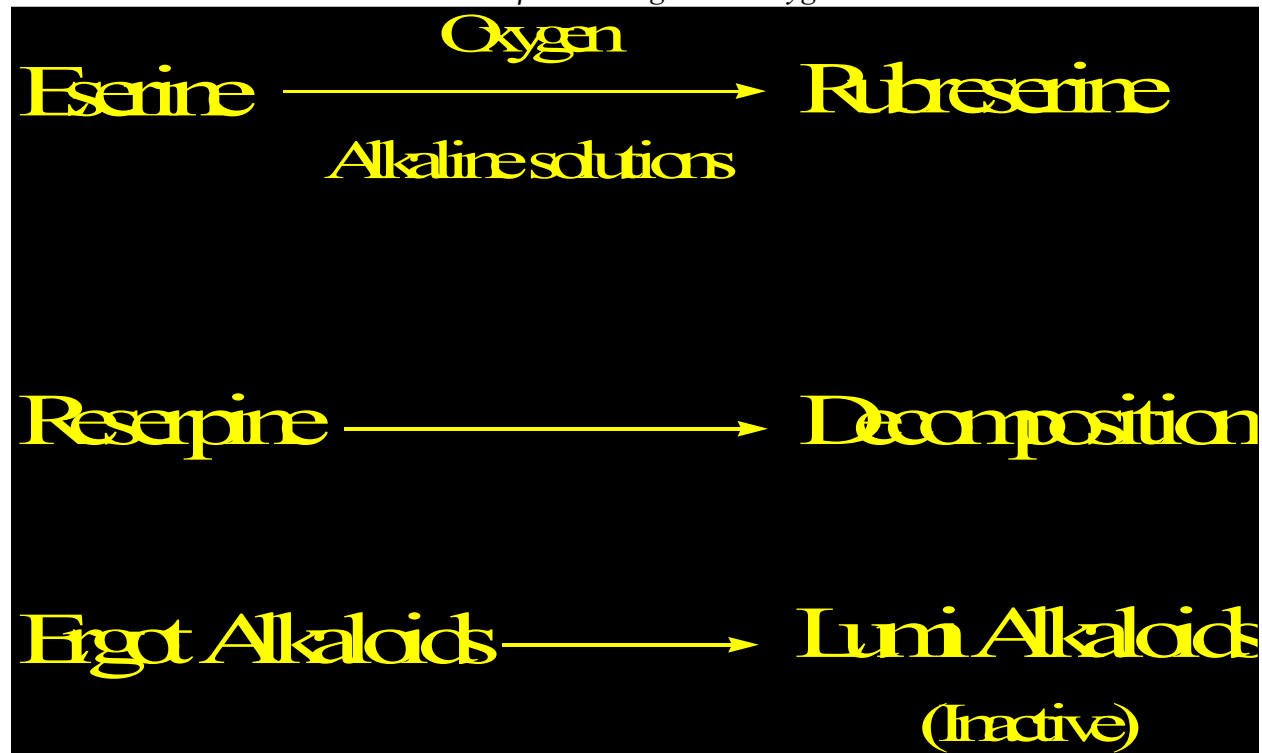
A. such as aqueous NaOH and KOH form salts with phenolic alkaloids

B. cause hydrolysis of ester alkaloids and amide alkaloids e.g. atropine, cocaine and physostigmine

C. cause opening of lactone ring

D. All

23. Some alkaloids are unstable when exposed to light and Oxygen: Which one is true?



D. All are correct

24. Which of the following is true about Extraction and purification of alkaloids?

A. 2 % Sulphuric acid extracts the alkaloids as their salts from powdered material

B. 70–80 % Ethanol or methanol: After extraction, acidified with dilute sulphuric acid and partitioned, then basified and re-extracted as above.

C.) Extraction with alkali treatment: The plant material is moistened with aqueous ammonia or calcium hydroxide which decomposes the alkaloid salts

D. All are correct

25. NH_4OH : Most widely used due to many advantages:

A. Strong enough to liberate most of alkaloids from their salts

B. Milder than fixed alkalis so more safe

c. Volatile so easy to get rid of it

D. All

26. Ether gives cleaner extract but have some disadvantages:

A. High volatility

B. Peroxide formation

C. Highly toxic

D. All

27. The best extraction for volatile alkaloids e.g. Nicotine is?

A. Acidify

B. Basicity

C. steam distillation

D. All are correct except C

28. Which of the following is true about Repeated Acid-Base procedures for purification of Crude Alkaloidal Fractions?

A. Render the extract Acidic, extract with organic solvent (dissolve non alkaloidal impurities)

B. Alkalinize and extract again with organic solvents (Dissolve Alkaloids).

C. Precipitation with alkaloidal precipitating agent.

D. Converted to crystalline salts.

E. All

28. Which of the following Classification of Alkaloids Based on the biogenetic pathway and precursor from which the alkaloids in question are produced

A. Taxonomic classification

C. Pharmacological classification

B. Biosynthetic classification

D. Chemical classification

29. ----- is most widely accepted and common mode of classification of alkaloids

A. Taxonomic classification C. Pharmacological classification

B. Biosynthetic classification **D. Chemical classification**

30. Which one is true about Biosynthesis of Alkaloids?

A. The biosynthesis of alkaloids starts from the primary metabolites of the amino acids or their decarboxylated products

B. most alkaloids are derived from α -amino acids

C. A couple of the β -amino acids are sometimes involved in alkaloid production also

D. All

31. Which one is β -amino acids?

A. nicotinic acid B. anthranilic acid C. A and B

32. Which of the following is not Ornithine Derived Alkaloids?

A. Necines B. Tropane C. Pyrrolidine D. None

33. Which of the following is pyrrolizidine Derived Alkaloids?

A. Necines B. Tropane C. Pyrrolidine D. None

34. Which of the following is not Lysine Derived Alkaloids?

A. piperidine B. quinolizidine C. indolizidine **D. None**

35. ----- are mainly found in plants of the Leguminosae/Fabaceae family and also known as lupine alkaloids

A. Ornithine Derived Alkaloids C. is pyrrolizidine Derived Alkaloids

B. quinolizidine alkaloids D. Lysine Derived Alkaloids

36. Which of the following is not quinolizidine alkaloids?

A. sparteine B. cytisine C. anagyrine **D. None**

37. Which of the following is indolizidine alkaloids?

A. swainsonine B. castanosperimine **C. A and B** D. None

38. Which of the following is true about Nicotinic Acid Derived Alkaloids?

A. Nicotinic acid is a non-protein amino acid

B. Biosynthesized with the condensation of aspartic acid and glyceraldehyde

C. Nicotinic acid is a vitamin B3 (niacin)

D. Contain pyridine ring

E. All

39. Which of the following is true about Alkaloids of tobacco?

A. Nicotine: a pyridine ring from nicotinic acid

B. Nicotine: a pyrrolidine ring from ornithine

C. Anabasin : a pyridine ring from nicotinic acid

D. .Anabasine: a piperidine ring from lysine

E. All

40. A large number of alkaloids are biosynthesized from tyrosine and they includes:

A. Simple phenylalkyl amines and tetrahydroisoquinolines

B. Colchicine

C. Emetine and related alkaloids

D. Tubocurarine and Opium alkaloids

E. All

41. Colchicine obtained from *Colchicum autumnale* (Liliaceae) and

A. is an effective treatment for acute gouty attacks

B. it is very toxic and not basic

C. causes teratogenic birth defects

D. All

42. ----- is the air-dried milky exudate, or latex, obtained by incising the unripe capsules of the opium poppy *Papaver somniferum* (*Papaveraceae*)?

A. Opium alkaloids

C. pyrrolizidine Derived Alkaloids

B. Ornithine Derived alkaloids

D. Lysine Derived Alkaloids

43. Which of the following is true about Opium Poppy: *Papaver somniferum*?

A. Member of the Papaveraceae, poppy family

B. Large colourfull annual with conspicuous flowers (white, pink, red, purple)

C. Fruit is a capsule that produces latex with several potent alkaloids

D. Latex is collected from cut capsules

E. All

44. Which of the following Opium alkaloids are used in medicine?

A. phenanthrenes

B. Benzylisoquinolines

C. A and B

45. Opium alkaloids usually occur in salt form combined with a unique acid known as meconic acid and

A. it occurs invariably in Papaver species and has not been detected in other plant species

B. it gives a deep red-coloured complex with ferric chloride

C. thus presence of meconic acid can be used to detect the presence of opium

D. This has been used as a rapid and reasonably specific test for opium

E. All

46. Which one is not simple indoles?

A. 5-Hydroxytryptamine (Serotonin) C. almotriptan

B. Sumatriptan D. eletriptan E. None

47 Which one is Ergot alkaloids or ergolines?

A.reserpine and deserpidine C. ajmalicine and ajmaline

B.rescinnamine and serpentine D. all

48. Which of the following is Cinchona alkaloids?

A. cinchonidine C. cinchonine

B. quinine D. quinidine E. All

49. Which of the following is Histidine Derived Alkaloids?

A. histamine C. Pilosine

B. pilocarpine D. All

50. Which of the following is Purine Alkaloids?

A. Caffeine

B. theobromine

C. theophylline

D. All

